RAJALAKSHMI ENGINEERING COLLEGE

Akash M 250701034-2025-CSE

A2

Started on	Friday, 25 September 2026, 3:30 PM
State	Finished
Completed on	Friday, 25 September 2026, 3:30 PM
Time taken	34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    int dp[n];

    // Read the sequence
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        dp[i] = 1;
    }

    int maxLength = 1;

    // Dynamic Programming
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++) {

            // Non-decreasing condition
            if (a[j] <= a[i]) {
                if (dp[j] + 1 > dp[i])
                    dp[i] = dp[j] + 1;
            }
        }

        if (dp[i] > maxLength)
            maxLength = dp[i];
    }

    printf("%d", maxLength);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course